

Discrete Mathematics

Chapter 1 The Foundations: Logic and Proofs

Logic = the study of correct reasoning

1. Propositions and Logical Operators

True value: $T(1)$ $F(0)$ Atom proposition is the simplest proposition.

proposition is a statement that is either true or false, but not both

Logical operators:

① Negation (not p): $\neg p$

if p is true, then $\neg p$ is false

② Conjunction (p and q): $p \wedge q$

if p and q are both true, then $p \wedge q$ is true

③ Disjunction (p or q): $p \vee q$

if p and q are both false, the $p \vee q$ is false

④ Implication (if p then q): $p \rightarrow q$

if p is true and q is false, then $p \rightarrow q$ is false

p : hypothesis q : conclusion

Everything implies truth; False implies anything.

⑤ Biconditional (p if and only if q): $p \leftrightarrow q$

if p and q have the same truth values, the $p \leftrightarrow q$ is true

Priority: $\neg$, then $\wedge$, $\vee$, then $\rightarrow$, $\leftrightarrow$

2. Propositional Formula

classification of propositional formula:

① tautology: if the truth table contains only true values for every case.

e.g. $p \rightarrow p \vee q$

② contradiction: if the truth table contains only false values for every case.

e.g. $p \wedge \neg p$

③ contingency : neither a tautology nor a contradiction

3. Propositional Equivalences

logically equivalent: if $A \Leftrightarrow B$ is tautology, denoted by $A \Leftrightarrow B$.

determine whether two propositions are logically equivalent is to use their truth tables.

$$\text{e.g. } \neg p \vee q \Leftrightarrow p \rightarrow q, \quad p \rightarrow q \Leftrightarrow \neg q \rightarrow \neg p, \quad (p \leftrightarrow q) \Leftrightarrow (p \rightarrow q) \wedge (q \rightarrow p) \Leftrightarrow (\neg p \vee q) \wedge (\neg q \vee p)$$

some important equivalences:

$$\text{identity laws} \quad p \vee F \Leftrightarrow p \quad p \wedge T \Leftrightarrow p$$

$$\text{domination laws} \quad p \vee T \Leftrightarrow T \quad p \wedge F \Leftrightarrow F$$

$$\text{idempotent laws} \quad p \vee p \Leftrightarrow p \quad p \wedge p \Leftrightarrow p$$

$$\text{complementation law} \quad \neg(\neg p) \Leftrightarrow p$$

$$\text{commutative law} \quad p \vee q \Leftrightarrow q \vee p \quad p \wedge q \Leftrightarrow q \wedge p$$

$$\text{associative law} \quad p \vee (q \vee r) \Leftrightarrow (p \vee q) \vee r \quad p \wedge (q \wedge r) \Leftrightarrow (p \wedge q) \wedge r$$

$$\text{distributive law} \quad p \wedge (q \vee r) \Leftrightarrow (p \wedge q) \vee (p \wedge r) \quad p \vee (q \wedge r) \Leftrightarrow (p \vee q) \wedge (p \vee r)$$

$$\text{law of excluded middle} \quad p \wedge \neg p \Leftrightarrow F \quad p \vee \neg p \Leftrightarrow T$$

$$\text{absorption law} \quad p \wedge (p \vee q) \Leftrightarrow p \quad p \vee (p \wedge q) \Leftrightarrow p$$

$$\text{De Morgan's law} \quad \neg(p \vee q) \Leftrightarrow \neg p \wedge \neg q \quad \neg(p \wedge q) \Leftrightarrow \neg p \vee \neg q$$

4. Propositional Normal Forms

① Disjunctive Normal Form : Conjunctive clauses and disjunctive normal form

in general, $(A_{11} \wedge A_{12} \wedge \dots \wedge A_{1n_1}) \vee \dots \vee (A_{k1} \wedge A_{k2} \wedge \dots \wedge A_{kn_k})$ only $\vee$ between two sentences

② Normal Forms

Theorem: Any formula A is tautologically equivalent to some formula is DNF.

③ Full Disjunctive Forms

minterm: each variable is represented exactly once (exist, and only once)

a boolean function is expressed as a disjunction of midterms.

$$\text{eg. } ((p \vee q) \rightarrow r) \rightarrow p \Leftrightarrow (\neg(p \vee q) \vee r) \rightarrow p \Leftrightarrow \neg(\neg(p \vee q) \vee r) \vee p$$

$$\Leftrightarrow (p \vee q) \wedge \neg r \vee p \Leftrightarrow (p \wedge \neg r) \vee (q \wedge \neg r) \vee p \Leftrightarrow p \vee (q \wedge \neg r)$$

Normal Forms are multiple.

if A and B are two different minterms, so $A \wedge B \equiv F$.

Use double negation and distributive laws to obtain full disjunctive forms ($A \Leftrightarrow A \vee T$)

eg. $p \vee (q \wedge \neg r) \Leftrightarrow [p \wedge (q \vee \neg q) \wedge (r \vee \neg r)] \vee [(q \wedge \neg r) \wedge (p \vee \neg p)]$

$\Leftrightarrow (p \wedge q \wedge r) \vee (p \wedge q \wedge \neg r) \vee (p \wedge \neg q \wedge r) \vee (p \wedge \neg q \wedge \neg r) \vee (\neg p \wedge q \wedge \neg r)$

$\Leftrightarrow m_2 \vee m_4 \vee m_5 \vee m_6 \vee m_7 \Leftrightarrow \sum (2, 4, 5, 6, 7)$

A is a tautology $\Leftrightarrow A = \sum_{i=0}^{2^N-1} (i)$

5. Proof

axiom: true without proof

theorem: statement shown to be true

lemma: a small theorem to prove bigger

corollary: a logical consequence of another theorem

conjective: truth value is unknown

$p \Rightarrow q, q \Rightarrow r$, then $p \Rightarrow r$

Rules of Inference:

Law of detachment: $p \rightarrow q, p$, therefore, q .

Modus tollens: $p \rightarrow q, \neg q$, therefore, $\neg p$.

Rule of addition: p , therefore, $p \vee q$.

Rule of simplification: $p \wedge q$, therefore, p .

Rule of conjunction: p, q , therefore, $p \wedge q$.

Hypothetical syllogism: $p \rightarrow q, q \rightarrow r$, therefore, $p \rightarrow r$.

Disjunctive syllogism: $p \vee q, \neg p$, therefore, q .

eg. $\neg p \wedge q, r \rightarrow p, \neg r \rightarrow s, s \rightarrow t \Rightarrow t$

① $\neg p \wedge q \Rightarrow \neg p$ ② $r \rightarrow p, \neg p \Rightarrow \neg r$ ③ $\neg r \rightarrow s, \neg r \Rightarrow s$ ④ $s \rightarrow t, s \Rightarrow t$.

eg. $p \rightarrow q, \neg p \rightarrow r, r \rightarrow s \Rightarrow \neg q \rightarrow s$

① $p \rightarrow q, \neg q \Rightarrow \neg p$ ② $\neg p \rightarrow r, \neg p \Rightarrow r$ ③ $r \rightarrow s, r \Rightarrow s$

If conclusion in form $p \rightarrow q$, we can convert the original problem to

$$p_1 \wedge p_2 \wedge \dots \wedge p_n \wedge p \Rightarrow q.$$

Assume p is true and q is false, show that $\neg p$ is also true.

$$\text{eg. } p \rightarrow (\neg(r \wedge s) \rightarrow \neg q), p, \neg s \Rightarrow \neg q$$

$$\textcircled{1} p, p \rightarrow (\neg(r \wedge s) \rightarrow \neg q) \Rightarrow \neg(r \wedge s) \rightarrow \neg q$$

$$\textcircled{2} \neg(\neg q), \neg(r \wedge s) \rightarrow \neg q \Rightarrow \neg\neg(r \wedge s) \Rightarrow r \wedge s \Rightarrow s$$

$$\textcircled{3} s, \neg s \Rightarrow \text{contradiction!}$$

$$\textcircled{4} \neg q.$$

b. Predicates and Quantifiers

predicates: a statement of the form $P(x_i)$ is the value of the propositional function

P at the n -tuple (x_i) . x_i is an element

quantifiers: For all x , $p(x)$: $\forall x, p(x)$

For some x , $p(x)$: $\exists x, p(x)$

$$\text{eg. } \forall \varepsilon > 0, \exists N > 0, \forall n > 0, |a_n - a| < \varepsilon. \text{ (definition of limit)}$$

$\forall, \exists$ can't be exchanged arbitrarily.

$$\lim_{x \rightarrow 0} f(x) = L \quad \forall \varepsilon > 0 \exists \delta > 0 \forall x (0 < |x - 0| < \delta \rightarrow |f(x) - L| < \varepsilon)$$

binding variables: when a quantifier is assigned to a variable

when a sentence has free variables, it will not be a statement

$$\text{tautology: } \forall x P(x) \vee \neg(\forall x P(x))$$

$$\text{contradictory: } \forall x P(x) \wedge \neg(\forall x P(x))$$

$$\text{contingence: } \forall x P(x) \rightarrow \exists y Q(y)$$

$$\text{De Morgan's laws: } \neg \forall x P(x) \Leftrightarrow \exists x \neg P(x) \quad \neg \exists x P(x) \Leftrightarrow \forall x \neg P(x)$$

$$\forall x (P(x) \wedge Q(x)) \Leftrightarrow (\forall x P(x)) \wedge (\forall x Q(x))$$

$$\exists x (P(x) \vee Q(x)) \Leftrightarrow (\exists x P(x)) \vee (\exists x Q(x))$$

$$\forall x (P(x) \vee Q(x)) \Leftrightarrow \forall x P(x) \vee \forall x Q(x)$$

$$\exists x (P(x) \wedge Q(x)) \Leftrightarrow \exists x P(x) \wedge \exists x Q(x)$$

$$\forall x Q(x) \rightarrow A \Leftrightarrow \exists x (Q(x) \rightarrow A)$$

$$\exists x Q(x) \rightarrow A \Leftrightarrow \forall x (Q(x) \rightarrow A)$$

7. Prenex Normal Form

$Q_1 x_1 Q_2 x_2 \dots Q_n x_n B$ (Q_i is $\forall$ or $\exists$ and B is quantifier free)

eg. $\forall x (\forall y P(y, x) \rightarrow \exists y Q(x, y))$

$$\Leftrightarrow \forall x (\neg \forall y P(y, x) \vee \exists y Q(x, y))$$

$$\Leftrightarrow \forall x (\exists y \neg P(y, x) \vee \exists z Q(x, z))$$

$$\Leftrightarrow \forall x \exists y \exists z (\neg P(y, x) \vee Q(x, z))$$

$$\Leftrightarrow \forall x \exists y \exists z (P(y, x) \rightarrow Q(x, z))$$

8. Rules of Inference for quantifiers

① Universal instantiation $\forall x \in D, P(x), d \in D$

therefore, $P(d)$

② Universal generalization $P(d)$ for any $d \in D$

therefore, $\forall x P(x)$

③ Existential instantiation $\exists x \in D, P(x)$

therefore $P(d)$ for some $d \in D$

④ Existential generalization $P(d)$ for some $d \in D$

therefore $\exists x P(x)$

eg. $\exists x (C(x) \wedge \neg B(x))$ and $\forall x (C(x) \rightarrow P(x))$

conc. $\exists x (P(x) \wedge \neg B(x))$.

① $\exists x (C(x) \wedge \neg B(x))$ ② $C(e) \wedge \neg B(e)$

③ $C(e)$ ④ $\forall x (C(x) \rightarrow P(x))$ ⑤ $C(e) \rightarrow P(e)$

⑥ $P(e)$ ⑦ $\neg B(e)$ ⑧ $P(e) \wedge \neg B(e)$

⑨ $\exists (P(x) \wedge \neg B(x))$

use existence first, then universal

eg. Proof $\forall x (P(x) \vee Q(x)) \Rightarrow \forall x P(x) \vee \exists x Q(x)$

$$\begin{aligned}
& \textcircled{1} \neg (\forall x P(x) \vee \exists x Q(x)) & \textcircled{2} \exists x \neg P(x) \wedge \forall x \neg Q(x) \\
& \textcircled{3} \exists x \neg P(x) & \textcircled{4} \neg P(e) & \textcircled{5} \forall x (P(x) \vee Q(x)) \\
& \textcircled{6} P(e) \vee Q(e) & \textcircled{7} Q(e) & \textcircled{8} \forall x \neg Q(x) \\
& \textcircled{9} \neg Q(e) & \textcircled{10} Q(e) \wedge \neg Q(e) \equiv F, \text{contradictory!}
\end{aligned}$$

Chapter 2 Basic Structures: Sets and Functions

1. Set

$|S|$ is the number of elements in S .

infinite countable sets: $\mathbb{Z}, \mathbb{Q}, \mathbb{N}, \dots$

uncountable: $\mathbb{R}, \mathbb{C}, \dots$

for any set A , $A \subseteq A$, $\emptyset \subseteq A \subseteq U$

proper subset: $\subset$ $A \subset B \Leftrightarrow A \subseteq B$ and $\exists x \in B, x \notin A$.

difference: $A - B = \{x \mid x \in A \wedge x \notin B\}$

complement: $\bar{A} = U - A$

symmetric difference: $A \oplus B = (A - B) \cup (B - A)$

$$A \oplus B = B \oplus A \quad (A \oplus B) \oplus C = A \oplus (B \oplus C) \quad (A \oplus A) = \emptyset \quad \emptyset \oplus A = A$$

power set: $2^S = \{T \mid T \subseteq S\}$.

Cartesian products: $A \times B = \{(a, b) \mid a \in A \wedge b \in B\}$.

$A^2 = A \times A$ (eg. $\mathbb{R}^2$)

if A and B are finite, $A \times B$ is finite, $|A \times B| = |A| |B|$

$$A \times \emptyset = \emptyset \times A = \emptyset, \quad A \times B \neq B \times A$$

$$A \times (B \cup C) = A \times B \cup A \times C, \quad A \times (B \cap C) = A \times B \cap A \times C$$

$$A_1 \times A_2 \times \dots \times A_n = \{(a_1, a_2, \dots, a_n) \mid a_i \in A_i\}$$

$$(A \cap B) \times (C \cap D) = (A \cap C) \times (B \cap D), \text{ but } (A \cup B) \times (C \cup D) \neq (A \cup C) \times (B \cup D)$$

cardinality of finite sets: $|S|$

$$|A \cup B| = |A| + |B| - |A \cap B|$$

cardinality of infinite sets:

if there is a one-to-one correspondence from A to B.

then A and B have the same cardinality.

countable infinite sets: $|A| = |\mathbb{N}|$

$$|\mathbb{N}| = |\mathbb{Z}|, |\mathbb{N}| = |\mathbb{Q}|, |\mathbb{N}^2| = |\mathbb{N}|$$

for any infinite set A, $|A| \geq |\mathbb{N}|$.

2. Function

one to one function: $f(a) = f(b) \Leftrightarrow a = b$

onto function: $\forall b \in D, \exists a \in I, f(a) = b$.

bijection function: correspondence

floor function: $\lfloor x \rfloor$ ceiling function: $\lceil x \rceil$

big-O notation: if $\lim_{x \rightarrow \infty} \frac{|f(x)|}{|g(x)|} = C, 0 < C < +\infty$.

we say $f(x)$ is $O(g(x))$.

if $f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0, (a_n \neq 0)$, Then $f(x)$ is $O(x^n)$.

Chapter 3 The Fundamentals: Algorithm

1. Algorithm

definition: a finite set of precise instruction for performing a computation

properties: ① input ② output ③ definiteness ④ correctness

⑤ finiteness ⑥ effectiveness ⑦ generality

2. Complexity

Best-time case and Worst-time case

Chapter 4 Induction and Recursion

1. Mathematical Induction

basic step $P(1)$ is shown to be true

inductive step $P(n) \rightarrow P(n+1)$ is shown to be true

2. Recursion

specify the value of function at zero

give a rule between larger number and smaller number

string is a finite sequence of symbols from alphabet Σ .

$$w \in \Sigma^*, x \in \Sigma \Rightarrow wx \in \Sigma^*$$

define $l(w)$ is the length of w , $l(xy) = l(x) + l(y)$

Chapter 5 Counting

1. Basic Rule

the sum rule: $n_1 + n_2 + \dots + n_m$ $|A_1 \cup A_2 \cup \dots \cup A_m| = |A_1| + |A_2| + \dots + |A_m|$

the product rule: $n_1 \cdot n_2 \cdot \dots \cdot n_k$ $|A_1 \times A_2 \times \dots \times A_n| = |A_1| |A_2| \dots |A_n|$.

2. The Pigeonhole Principle

if $k+1$ or more objects and k boxes, then at least one box contains two or more.

generalized: N objects and k boxes, the $\lceil \frac{N}{k} \rceil$.

Theorem: Every sequence of n^2+1 distinct real numbers contains a subsequence of length that is either strictly increasing or decreasing.

proof: let i_k be the maximum of the length of increasing sequences,

d_k be the maximum of the length of decreasing sequences.

suppose there's no subsequence that satisfies the need,

then $1 \leq i_k, d_k \leq n$, so there are n^2 possible pairs,

$$\Rightarrow \exists s < t < n^2+1, (i_s, d_s) = (i_t, d_t),$$

if $a_s < a_t$, then $i_t > i_s$; $a_s > a_t$, then $d_s > d_t$.

so ...

3. Permutation and Combination

$$P(n, r) = n(n-1) \dots (n-r+1), \quad C(n, r) = P(n, r) / r!$$

4. Generalized Permutation and Combination

r objects' permutation with repetition: n^r

r objects' combination with repetition: $C(n+r-1, r)$

5. Generating Permutation and Combination

find the decreasing subsequence $a_{j+1}, \dots, a_n$

then replace a_j with the least number that is larger than it.

Chapter 6 Advanced Counting Techniques

Chapter 6 Advanced Counting Techniques

1. Recurrence Relations

a_n is in terms of previous terms

linear homogeneous recurrence relation:

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k}, \text{ assume } a_n = r^n.$$

characteristic equation: $r^k - c_1 r^{k-1} - \dots - c_k = 0$

general solutions: $a_n = \alpha_1 r_1^n + \alpha_2 r_2^n + \dots + \alpha_k r_k^n$, α_i are consts.

(also see the ordinary differential equation).

need initial values to confirm α_i .

if r is k -th root, then $a_n = (\alpha_1 + \alpha_2 n + \dots + \alpha_{k-1} n^{k-1}) r^n$.

nonhomogeneous recurrence relation:

a particular solution + general solutions

eg. $a_n = 3a_{n-1} + 2n$

$$\text{let } a_n = cn + d \Rightarrow cn + d = 3[c(n-1) + d] + 2n$$

$$c = -1, d = -\frac{3}{2}, a_n = -3^n - \frac{1}{2}n - \frac{3}{2}.$$

(also see my "ordinary differential equations" notes)

2. Generating Functions

a number sequence $\{a_n\}$, generating function:

$$G(x) = a_0 + a_1 x + \dots + a_n x^n + \dots = \sum a_n x^n.$$

its calculating properties are like power functions

(also see my "mathematical analysis" notes)

extended binomial coefficient: $\binom{u}{k} = u(u-1)(u-2)\dots(u-k+1)/k!$

$$\Rightarrow (1+x)^u = \sum \binom{u}{k} x^k$$

$$(1+x)^{-n} = \sum (-1)^k \binom{n+k-1}{k} x^k$$

eg. $a_k = 3a_{k-1}$, $G(x) = \sum a_n x^n$

$$a_k x^k = 3a_{k-1} x^k \Rightarrow \sum a_k x^k = \sum 3a_{k-1} x^k$$

$$\Rightarrow G(x) - a_0 = 3xG(x) , \quad G(x) = \frac{a_0}{1-3x}$$

using Taylor expansion, we can work out $G(x)$

eg. $x_1 + x_2 + x_3 = 7$, but $2 \leq x_1 \leq 5$, $3 \leq x_2 \leq 6$, $4 \leq x_3 \leq 7$

$$(x^2 + x^3 + x^4 + x^5)(x^3 + x^4 + x^5 + x^6)(x^4 + x^5 + x^6 + x^7)$$

the coefficient of x^7 .

3. Applications of Inclusion-Exclusion

$$|\bigcup_{i=1}^n A_i| = \sum |A_i| - \sum_{1 \leq i < j \leq n} |A_i \cap A_j| + \dots + (-1)^{n+1} |\bigcap_{i=1}^n A_i|$$

Chapter 7 Relations

1. Definition: A binary relation between A and B, $R \subseteq A \times B$.

relations between set A and B have $2^{|A||B|}$ possibilities.

composite: $S \circ R = \{(a, c) \mid a \in A, c \in C, \exists b: (a, b) \in R, (b, c) \in S\}$.

inverse: $R^{-1} = \{(y, x) \mid (x, y) \in R\}$

reflexive: $\forall x \in A, (x, x) \in R$ ($\leq, \geq$)

irreflexive: $\forall x \in A, (x, x) \notin R$ ($<, >$)

R is reflexive $\Leftrightarrow R = \subseteq R$ ($2^{n(n-1)}$ possibilities)

R is irreflexive $\Leftrightarrow R = \cap R = \emptyset$. ($2^{n(n-1)}$ possibilities)

symmetric: $\forall x, y \in A, (x, y) \in R \Rightarrow (y, x) \in R$

antisymmetric: $\forall x, y \in A, (x, y) \in R \wedge (y, x) \in R \Rightarrow x = y$

R is symmetric $\Leftrightarrow R = R^{-1}$ ($2^{\frac{n(n+1)}{2}}$ possibilities)

R is antisymmetric $\Leftrightarrow R \cap R^{-1} \subseteq R =$ ($2^n \cdot 3^{\frac{n(n-1)}{2}}$ possibilities)

$\{(1,1)\}$ is both symmetric and antisymmetric

transitive: $\forall x, y, z \in A, (x,y) \in R \wedge (y,z) \in R \Rightarrow (x,z) \in R$

R is transitive $\Leftrightarrow R \circ R \subseteq R$

Theorem: R is transitive $\Leftrightarrow R^n \subseteq R$ for any n .

2. Representing Relations

matrices of relations: $A = \{a_m\}, B = \{b_n\}$

$M_R = (m_{ij})_{m \times n}$ and $m_{ij} = 1$ if $(a_i, b_j) \in R$

$m_{ij} = 0$ if $(a_i, b_j) \notin R$

R is reflexive $\Leftrightarrow m_{ii} = 1$

R is symmetric $\Leftrightarrow M_R = M_R^T$

R is transitive $\Leftrightarrow$ if c_{ij} in M_R^2 is nonzero then m_{ij} in M_R is nonzero

$$M_{R_1 \cup R_2} = M_{R_1} \vee M_{R_2}$$

$$M_{R_1 \cap R_2} = M_{R_1} \wedge M_{R_2}$$

$$M_{S \circ R} = M_R \odot M_S$$

digraphs of relations

R is reflexive $\Leftrightarrow$ loops at every vertex

R is symmetric $\Leftrightarrow$ has both directions

3. Closure of Relations

Definition: R is a relation, S with property P , and $S \supseteq R$

$\forall S'$ with property $P, S' \supseteq R \Rightarrow S \subseteq S'$

Then S is called the closure of R with respect to P .

Theorem: The reflexive closure of relation R :

$$r(R) = R \cup \Delta, \Delta = \{(a,a), (b,b), \dots\}$$

The symmetric closure of relation R :

$$S(P) = R \cup R^{-1}$$

Theorem: There is a path of length n from a to $b \Leftrightarrow (a,b) \in R^n$.

connective relation: $R^* = \{(a,b) \mid \text{there is a path from } a \text{ to } b\}$

$$R^* = \bigcup_{n=1}^{\infty} R^n$$

Theorem: The transitive closure of R is

$$t(R) = R^*.$$

Theorem: If M_R is a 0-1 matrix of the relation R on a set with n elements. Then the matrix of the transitive closure R^* is

$$M_{R^*} = M_R \vee M_R^2 \vee \dots \vee M_R^n.$$

Warshell's Algorithm

$$w_{ij}^{[k]} = w_{ij}^{[k-1]} \vee (w_{ik}^{[k-1]} \wedge w_{kj}^{[k-1]})$$

4. Equivalence Relations

Definition: $R \sim$ if reflexive, symmetric, transitive

Equivalence Classes: $[a]_R = \{x \in A \mid (x,a) \in R\}$

If $b \in [a]_R$, b is a representative.

1) $\forall a \in A, [a]_R \neq \emptyset$.

2) $(a,b) \in R \Rightarrow [a]_R = [b]_R$

3) $(a,b) \notin R \Rightarrow [a]_R \cap [b]_R = \emptyset$

Quotient Set: $A/R = \{[a]_R \mid a \in A\}$

Partition: $\bigcup_{k=1}^n A_k = S$, $A_j \cap A_i = \emptyset$ for all $i \neq j$.

Theorem: If R is an equivalence relation, then the partitions of R form a partition of X .

And given a partition, there is a corresponding equivalence relation.

5. Partial Ordering

Definition: $R \leq$, if reflexive, antisymmetric, transitive

Comparable: If for $a, b \in (S, \leq)$, either $a \leq b$ or $a \geq b$

If for all $a, b \in S$, a, b is comparable, then S is a totally order.

Lexicographic Order: $(a_1, a_2) < (b_1, b_2)$

either $a_1 < b_1$ or $a_1 = b_1 \wedge a_2 < b_2$.

Maximal/Minimal Element: $a \in B \wedge$ There is no x that $x > a / x < a$.

Greatest/Least Element: $a \in B \wedge \forall x \in B, x \leq a / x \geq a$

Upper/Lower Bound: $c \in A \wedge \forall x \in B, x \leq c / x \geq c$.

Least Upper / Greatest Lower Bound: Similar to mathematical analysis.

Remark: maximal/minimal element is not unique

greatest/least element can have at most one

Well-ordered Set: $(S, \leq)$ is a total ordering and has a least element

Lattice: A partially ordered set in which pair of element has both a least upper bound and a greatest lower bound.

Chapter 8 Graphs

1. Graph

Simple Graph: $G = (V, E)$ $V \neq \emptyset$, E is unordered

Multigraph: $E: \{u, v\}, u \neq v$, multi edges, but loops are not allowed

Pseudograph: Allow loops

Directed Graph: E is ordered

Directed Multigraph: Multi edges, allow loops

2. Graph Terminology

Adjacent / Neighbors: $e = \{u, v\}$ is an edge

$e = \{u, v\}$, e is called incident with u and v = endpoints

Degree: numbers of edges incident with it, loop counts twice

Theorem: G is a graph with m edges and n vertices, then

$$\sum_{i=1}^n \deg(v_i) = 2m$$

Directed Graphs: (u, v) u : initial vertex v : end vertex

in-degree: ^-v , numbers of edges with v as their terminal vertex

out-degree: ^+v , ... initial vertex

Theorem: $\sum \deg^-(v_i) = \sum \deg^+(v_i) = |E|$

Complete Graph: K_n is a simple graph with n vertices and each pair is joined by an edge.

$$|E| = C_n^2.$$

Cycles: $\{v_1, v_2\}, \{v_2, v_3\}, \dots, \{v_n, v_1\}$.

Wheels: Cycles with a center point and edges.

Bipartite Graph: $V = V_1 \cup V_2, V_1 \cap V_2 = \emptyset$.

No edges exist in the same subset.

Complete Bipartite Graph

$$G = (V, E), H = (W, F)$$

subgraph: $W \subseteq V, F \subseteq E$

spanning subgraph: $W = V, F \subseteq E$

3. Representing Graphs

Adjacency Matrix: For a simple graph, $a_{ij} = 1$ if $\{v_i, v_j\} \in E$

1) Symmetric

2) $a_{ii} = 0$

3) $\deg(v_i) = \sum_{j=1}^n a_{ij}$

Adjacency Matrix: For a pseudograph, a_{ij} = the numbers of edges that associated with $\{v_i, v_j\}$

Incidence Matrix: $m_{ij} = 1$ when edge e_j is incident with v_i

Isomorphic: If there's a bijection $f: V_1 \rightarrow V_2, \forall a, b \in V_1, (a, b) \in E_1$

$$\Leftrightarrow (f(a), f(b)) \in E_2.$$

4. Connectivity

Path: $u \rightarrow v$, $f(e_1) = \{x_0, x_1\}$, $f(e_2) = \{x_1, x_2\}$, $\dots$, $f(e_n) = \{x_{n-1}, x_n\}$

$$x_0 = u, x_n = v$$

Theorem: There are a simple path between every pair of distinct vertices of a connected undirected graph.

Cut vertices: If you delete such vertices, the graph is not connected anymore.

Cut edges/Bridges

Strongly connected: A path from $a \rightarrow b$ and $b \rightarrow a$

Weakly connected: Connected in the underlying undirected graph.

Theorem: A is the adjacency matrix of a graph G , the number of different paths of length r from v_i to v_j equals to the (i, j) th entry of A^r .

5. Euler and Hamilton Paths

Euler circuit: A simple circuit contains every edge of G . (Euler graph)

Euler path: A simple path contains every edge of G .

Theorem: A connected multigraph has an Euler circuit

$\Leftrightarrow$ Every vertex in V has even degree.

Hamilton circuit: A circuit which visits every vertex in V exactly once.

Theorem: If G is a connected simple graph with n vertices where $n \geq 3$, then G has a Hamilton circuit if the degree of each vertex is at least $\frac{n}{2}$.

If for $v_i, v_j \in V$, $\deg(v_i) + \deg(v_j) \geq n$, then G is a Hamilton graph.

6. Shortest Path Problems

Dijkstra's algorithm: $O(n^2)$

7. Planar Graphs

Definition: If a graph can be drawn in the plane without crossing edges.

Region: A planar representation of a graph into regions, including an unbounded region.

Euler's Formula: G is a connected planar simple graph with e edges, v vertices.

r is the number of regions,

Then $v - e + r = 2$. (Proved by induction of e)

If G is an undirected planar graph, with p connected components, the $v - e + r = p + 1$.

Corollary: If G is a connected planar simple graph with e edges and v vertices

where $v \geq 3$, then $e \leq 3v - 6$.

Degree of region: The number of edges on the boundary of region.

$$\Rightarrow \sum \deg(r_i) = 2e.$$

In simple graph, $\deg(r_i) \geq 3$

Corollary: If there's no circuit of length 3, then $e \leq 2v - 4$.

Kuratowski's Theorem:

Elementary Subdivision: Let $\{u, v\} \in E$, remove edge $\{u, v\}$, adding w and $\{u, w\}, \{w, v\}$.

A graph G is nonplanar if and only if G contains a subgraph homeomorphic to K_5 or $K_{3,3}$.

8. Graph Coloring

The Four Color Theorem:

The chromatic number of a planar graph is no greater than four.

- Only available for planar graph.

chromatic number of $K_n = n$, $K_{m,n} = 2$.

$$C_{2k} = 2, C_{2k+1} = 3$$

Chapter 9 Trees

1. Tree

Tree: Connected undirected graph with no simple circuit.

Theorem: An undirected graph is a tree iff a unique path connects every two vertices.

Root: A particular vertex of a tree.

parent, child, siblings, ancestor, descenstor

Leaf: Vertices that have no children.

Level: $l(v)$ the length of the simple path from root to v

Height: $h = \max_{v \in V(T)} l(v)$

m-ary tree: every internal vertex has no more than m children

full tree: has exactly m children

Balanced: All leaves are at levels of h or $h-1$

Theorem: A tree with n vertices has $n-1$ edges.

Theorem: A full m -ary tree with i internal vertices contains $n = mi+1$ vertices.

Theorem: There are at most m^h leaves in m -ary tree of height h .

Corollary: If an m -ary tree of height h has l leaves, then $h \geq \lceil \log_m l \rceil$

2. Application

Binary Search Tree

Prefix Code

3. Spanning Trees

Definition: T is a spanning tree of G , if

T is a sub graph of G , T contains all the vertices of G .

Theorem: A simple graph is connected if and only if it has a spanning tree.

Minimum Spanning Tree: With the smallest possible sum of weights of its edges.

§ Final Review

1. Propositional Logic

① Propositions and Logical Operators $\{\neg, \wedge, \vee, \rightarrow, \leftrightarrow\}$

② Formulas $\{\text{Tautology, Contradiction, Contingence}\}$

③ Truth table, Logical equivalence, Normal forms

④ Propositional Equivalence

⑤ Normal Forms Conjunctive and Disjunctive Normal Form

⑥ Law of Argument

⑦ Predicates and Quantifiers $\forall, \exists$

⑧ Logical Equivalence $\forall x(p(x) \wedge q(x)) \Leftrightarrow \forall x p(x) \wedge \forall x q(x)$

$$\exists x(p(x) \vee q(x)) \Leftrightarrow \exists x p(x) \vee \exists x q(x)$$

2. Set Theory

① Set Operations

② Power Set $P(S) = 2^S$

③ Cartesian Products $A \times B = \{(a, b) \mid a \in A, b \in B\}$

④ Cardinality of Finite Sets inclusion-exclusion

⑤ Infinite Sets countable

3. Relation

① Reflexive, Symmetric, Transitive

② Closure of Relations

③ Equivalence Relation

④ Partial Ordering (Antisymmetric)

Total Ordering, Hasse Diagram, Elements

Lattice (Algebra System)

4. Counting

① Counting Principle

② Pigeonhole Principle

③ Permutations and Combination

④ Recurrence

⑤ Generating Functions

5. Graphs and Trees